

Mathematical Foundations of Data Science

Concepts, Intuition, and Simple Calculations

BSIT LECTURE HANDOUT

Course: Data Science Program: BSIT

Numbers • Vectors • Probability • Statistics • Optimization

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1 Why Mathematics Matters in Data Science

Mathematical Foundations

Mathematical foundations of data science are the core ideas from arithmetic, algebra, linear algebra, probability, statistics, and calculus that help us describe data, build models, measure uncertainty, and optimize predictions.

Big Picture

Data science is not only about coding tools. Every chart, model, prediction, and evaluation metric rests on mathematical reasoning. Mathematics helps answer questions such as: How large is the average? How uncertain is a prediction? How similar are two records? How does a model improve step by step?

1.1 What Mathematics Helps Us Do

- Summarize datasets using averages and spread.
- Represent data in vectors and matrices.
- Measure uncertainty using probability.
- Find relationships using correlation and covariance.
- Minimize errors using optimization.

1.2 A Quick Roadmap

Math Area	Common Data Science Use
Algebra	Writing formulas for models and metrics
Linear algebra	Storing features and performing transformations
Probability	Modeling uncertainty and event likelihood
Statistics	Summarizing, comparing, and interpreting data
Calculus	Improving models by minimizing loss functions

2 Numbers, Algebra, and Functions

2.1 Basic Arithmetic in Data Science

Data science often begins with basic operations:

- Addition and subtraction for totals and changes.
- Multiplication and division for rates and averages.
- Ratios and percentages for comparison.

Percentage Example

Suppose 18 out of 24 students passed a quiz. The pass rate is

$$\text{Pass rate} = \frac{18}{24} \times 100 = 75\%$$

This simple percentage is already a useful data summary.

2.2 Variables and Simple Formulas

In mathematics, a variable is a symbol that represents a value. For example, if x is the number of hours studied, then a very simple score model might be

$$y = 5x + 40$$

If a student studies for $x = 6$ hours, then

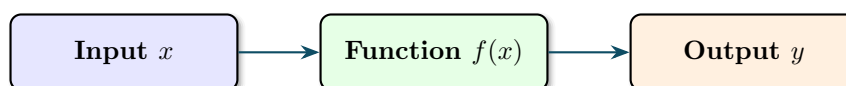
$$y = 5(6) + 40 = 70$$

2.3 Functions

A function maps an input to an output.

Function

A function is a rule that takes one or more inputs and produces an output. In data science, models are functions that transform input features into predictions.



2.4 Common Function Shapes

- **Linear:** $y = mx + b$
- **Quadratic:** $y = ax^2 + bx + c$
- **Exponential:** $y = ab^x$
- **Sigmoid:** $\sigma(z) = \frac{1}{1+e^{-z}}$

Why Functions Matter

Regression, classification, recommendation systems, and neural networks all rely on functions. The main difference is how simple or complex the function is.

3 Vectors and Matrices

3.1 Vectors

A vector is an ordered list of numbers. In data science, one row of features can be represented as a vector.

$$\mathbf{x} = \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}$$

If the values represent *hours studied*, *assignments completed*, and *attendance score*, then the vector stores one student's features.

</> Vector Addition

Let

$$\mathbf{a} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$$

Then

$$\mathbf{a} + \mathbf{b} = \begin{bmatrix} 1 + 3 \\ 2 + 5 \end{bmatrix} = \begin{bmatrix} 4 \\ 7 \end{bmatrix}$$

3.2 Dot Product

The dot product combines two vectors into one number:

$$\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2 + \cdots + a_nb_n$$

</> Dot Product Calculation

Let

$$\mathbf{a} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

Then

$$\mathbf{a} \cdot \mathbf{b} = (2)(4) + (3)(5) = 8 + 15 = 23$$

The dot product is important in linear models and similarity calculations.

3.3 Matrices

A matrix is a rectangular table of numbers. Datasets are often stored as matrices.

$$X = \begin{bmatrix} 1 & 2 & 80 \\ 2 & 4 & 75 \\ 3 & 5 & 92 \end{bmatrix}$$

Each row can represent one observation, and each column can represent one feature.

💡 Matrix Interpretation

If a dataset has m rows and n columns, then it can be seen as an $m \times n$ matrix. Many machine learning algorithms operate directly on this structure.

3.4 Distance Between Two Points

For two points (x_1, y_1) and (x_2, y_2) , Euclidean distance is

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

</> Distance Example

Find the distance between $(1, 2)$ and $(4, 6)$:

$$d = \sqrt{(4 - 1)^2 + (6 - 2)^2} = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$$

Distance is used in clustering and nearest-neighbor methods.

4 Statistics for Describing Data

4.1 Mean, Median, and Mode

Suppose the quiz scores are: 60, 70, 70, 80, 90

- **Mean:** average value
- **Median:** middle value after sorting
- **Mode:** most frequent value

Mean, Median, Mode

For the scores 60, 70, 70, 80, 90:

$$\text{Mean} = \frac{60 + 70 + 70 + 80 + 90}{5} = \frac{370}{5} = 74$$

Sorted data is already 60, 70, 70, 80, 90, so

$$\text{Median} = 70$$

The most frequent value is also

$$\text{Mode} = 70$$

4.2 Range, Variance, and Standard Deviation

- **Range:** maximum minus minimum
- **Variance:** average squared distance from the mean
- **Standard deviation:** square root of the variance

Variance and Standard Deviation

Consider the data: 2, 4, 6

Step 1: Find the mean.

$$\bar{x} = \frac{2 + 4 + 6}{3} = 4$$

Step 2: Find deviations from the mean.

$$2 - 4 = -2, \quad 4 - 4 = 0, \quad 6 - 4 = 2$$

Step 3: Square the deviations.

$$(-2)^2 = 4, \quad 0^2 = 0, \quad 2^2 = 4$$

Step 4: Average the squared deviations.

$$\text{Variance} = \frac{4 + 0 + 4}{3} = \frac{8}{3} \approx 2.67$$

Step 5: Take the square root.

$$\text{Standard deviation} = \sqrt{\frac{8}{3}} \approx 1.63$$

4.3 Why Spread Matters

Two datasets can have the same mean but different variability. In practice, a low-spread dataset is more consistent, while a high-spread dataset is more scattered.

⚠ Practical Warning

An average alone can be misleading. If two classes both have mean score 75, but one class has very large spread, then performance consistency is quite different.

4.4 Z-Score

A z-score shows how far a value is from the mean in standard deviation units:

$$z = \frac{x - \bar{x}}{s}$$

🔗 Z-Score Example

Suppose a student score is $x = 85$, the class mean is $\bar{x} = 75$, and the standard deviation is $s = 5$.

$$z = \frac{85 - 75}{5} = 2$$

This means the score is 2 standard deviations above the mean.

5 Probability and Uncertainty

5.1 Basic Probability

Probability measures how likely an event is.

$$P(A) = \frac{\text{Number of favorable outcomes}}{\text{Total number of outcomes}}$$

🔗 Simple Probability Example

If a bag contains 3 red balls and 2 blue balls, then the probability of selecting a red ball is

$$P(\text{red}) = \frac{3}{5} = 0.6$$

So there is a 60% chance of selecting red.

5.2 Conditional Probability

Conditional probability asks for the probability of one event given that another event has already happened.

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

🔗 Conditional Probability Example

Suppose 20 students are in a class. 12 are male, and 5 are male and play basketball. Then the probability that a randomly chosen student plays basketball given that the student is male is

$$P(\text{basketball} | \text{male}) = \frac{5}{12}$$

5.3 Bayes' Idea

Bayes' theorem updates probability when new evidence arrives:

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)}$$

💡 Why Bayes Matters

Spam detection, medical diagnosis, fraud detection, and classification all use the idea of updating beliefs when evidence is observed.

5.4 Expected Value

Expected value is the long-run average outcome:

$$E(X) = \sum xP(x)$$

🔗 Expected Value Example

Suppose a simple game pays 10 pesos with probability 0.3 and 0 pesos with probability 0.7.

$$E(X) = 10(0.3) + 0(0.7) = 3$$

The expected value is 3 pesos.

6 Covariance, Correlation, and Relationships

6.1 Covariance

Covariance tells us whether two variables tend to move together.

- Positive covariance: both increase together.
- Negative covariance: one increases while the other decreases.
- Near zero covariance: weak linear movement together.

6.2 Correlation

Correlation standardizes covariance so that the result lies between -1 and 1 .

- $r = 1$: perfect positive linear relationship
- $r = -1$: perfect negative linear relationship
- $r = 0$: no linear relationship

🔗 Interpreting Correlation

If study hours and exam score have correlation $r = 0.82$, that suggests a strong positive linear relationship. As study hours increase, scores also tend to increase.

⚠ Important Caution

Correlation does not prove causation. Two variables can move together without one directly causing the other.

7 Calculus and Optimization Intuition

7.1 Slope and Derivative

The derivative tells us how fast a quantity changes.

📖 Derivative

The derivative measures the rate of change of a function. In machine learning, derivatives help determine how to adjust parameters to reduce error.

For example, if

$$f(x) = x^2$$

then

$$f'(x) = 2x$$

🔗 Simple Derivative Example

For $f(x) = x^2$, find the slope at $x = 3$:

$$f'(x) = 2x$$

So at $x = 3$,

$$f'(3) = 2(3) = 6$$

This means the curve is increasing with slope 6 at that point.

7.2 Loss Function

A loss function measures prediction error. One common example is mean squared error:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

🔗 MSE Calculation

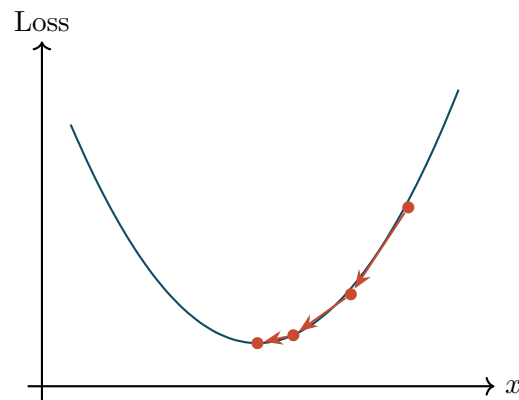
Suppose actual values are $[4, 6]$ and predicted values are $[5, 5]$.

$$\text{MSE} = \frac{(4 - 5)^2 + (6 - 5)^2}{2} = \frac{1 + 1}{2} = 1$$

The average squared error is 1.

7.3 Gradient Descent Intuition

Gradient descent is a step-by-step method for reducing error.



💡 Optimization Idea

If a model makes large errors, gradient-based optimization changes the parameters in the direction that reduces the loss. This is one of the key ideas behind linear regression, logistic regression, and neural networks.

8 From Mathematics to Machine Learning Models

8.1 Linear Regression Idea

Linear regression predicts a numeric output using a straight-line equation:

$$\hat{y} = b_0 + b_1x$$

🔗 Linear Regression Example

Suppose a model is

$$\hat{y} = 40 + 5x$$

If a student studies for $x = 7$ hours, then the predicted score is

$$\hat{y} = 40 + 5(7) = 75$$

8.2 Logistic Regression Idea

Logistic regression predicts probabilities for two-class problems using the sigmoid function:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

🔗 Probability Interpretation

If a logistic model outputs 0.82, that means the model estimates an 82% probability that the observation belongs to the positive class.

8.3 Feature Scaling

Features measured on very different scales can distort calculations.

🔗 Scaling Example

Suppose one feature is *age* with values around 20, and another is *family income* with values around 50000. The income feature may dominate distance or gradient calculations.

Standardization helps by putting features on comparable scales.

9 Quick Reference Summary

Concept	Key Formula or Idea
Mean	$\bar{x} = \frac{\sum x}{n}$
Variance	Average squared distance from the mean
Standard deviation	Square root of variance
Probability	$P(A) = \frac{\text{favorable}}{\text{total}}$
Distance	$d = \sqrt{\sum (x_i - y_i)^2}$
Dot product	$\mathbf{a} \cdot \mathbf{b} = \sum a_i b_i$
Derivative	Rate of change of a function
MSE	$\frac{1}{n} \sum (y_i - \hat{y}_i)^2$
Correlation	Strength and direction of linear relationship

10 Key Takeaways and Review Questions

10.1 Key Takeaways

- Data science relies on core mathematics, not just software tools.
- Algebra and functions help express models and predictions.
- Vectors and matrices organize data for machine learning.
- Statistics summarizes center and spread.
- Probability handles uncertainty and evidence.
- Calculus supports optimization and model improvement.

10.2 Review Questions

1. What is the difference between a mean and a median?
2. How is a vector different from a matrix?
3. Why is probability important in classification problems?
4. What does a positive correlation mean?
5. Why do machine learning algorithms use loss functions?
6. What role does the derivative play in optimization?

Short Reflection Task

Choose a small dataset from your daily life, such as quiz scores, phone usage hours, or daily expenses. Compute the mean, range, and standard deviation. Then explain what those numbers reveal about the data.

End of Handout

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